

generalised Fermat curve

$$C = X_{d,1}(\lambda) \subset \mathbb{P}^N$$

$C \rightarrow \mathbb{P}^1$ Galois, with group

$$H = \mu_d^{N+1} / \mu_d, \text{ of genus}$$

$$1 + \frac{1}{2}d^{N-1}((N-1)(d-1) - 2)$$



the complete intersection

$$X \cong X_{d,r}(\lambda)$$

$c = N - r$ diagonal equations

$$\Phi(u)_i = \prod_j u_i^{(j)}, X \cong C^r / G$$

$$\deg \Phi = |G| = r! d^{N(r-1)}$$

$X \in \mathcal{A}$: (F3') in every degree



the Hodge classes of degree r

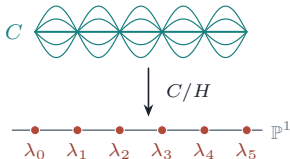
$$H^r(X) = \mathbb{C}\eta \oplus \bigoplus_{[a]} W_{[a]}$$

$$W_{[a]} = \bigoplus_{a' \in [a]} \bigwedge^r V_{a'}$$

$$\dim V_a = \#\{i : a_i \neq 0\} - 2$$

$W_{[a]}$ is the image of $H^r(B_{[a]})$

under an algebraic correspondence



known: $\dim B_{[a]} \leq 5$

Markman, Moonen-Zarhin

$d \in \{3, 4, 6\}$ and $N \leq 6$,

or $d = 2$ and $N \leq 12$;

70, 490, 6125 Weil orbits in $\mathbb{P}^6$

open: $\dim B_{[a]} > 5$

$N \geq 7$ and $d \in \{3, 4, 6\}$,

or $N \geq 13$ and $d = 2$

a case of (F2)

open: $\varphi(e) \geq 4$

$d = 5$ or $d \geq 7$

Weil classes of $\mathbb{Q}(\zeta_e)$

$P2_{\text{split}}$ of the paper

very general λ , every N : the Hodge conjecture holds for $d = 2$ in every dimension, for $d = 3, 4$ up to dimension 7

and for $d = 6$ up to dimension 5, through the monodromy of the cyclic covers of degree 3, 4 and 6 of $\mathbb{P}^1$